# Project 01: AI customer ticket classifier

## Business Questions & Findings


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## Contents

## Q1) what are customers contacting us about most often?

For this project, eight banking-support intents were selected from the BANKING77 dataset.

The training data shows that the most frequent selected intents are:

- cash_withdrawal_charge: 177 tickets
- transaction_charged_twice: 175 tickets
- transfer_not_received_by_recipient: 171 tickets
- cash_withdrawal_not_recognised: 160 tickets
- card_arrival: 153 tickets
- declined_card_payment: 153 tickets
- card_not_working: 112 tickets
- lost_or_stolen_card: 82 tickets

Therefore, among the selected categories, cash-withdrawal charges, duplicate transactions, and transfer-related issues represent the largest portions of the workload.

The distribution is not perfectly balanced, but the differences are moderate. The test set was intentionally balanced with 40 samples per intent, allowing the model to be evaluated equally across all eight categories.

Business implication:

Higher-volume categories may deserve greater operational attention when designing automated routing or support workflows.

## Q2) do message length and structure differ by intent?

Message length was analyzed using both word count and character count.

The median word count across the selected intents ranges from approximately 9 to 12 words. "transfer_not_received_by_recipient" and "cash_withdrawal_not_recognised" tend to have longer messages, while "card_not_working", "declined_card_payment", and "lost_or_stolen_card" tend to have shorter messages.

Character length shows somewhat clearer differences between some intents. However, there is still considerable overlap between categories.

Overall, message length provides some descriptive information about the different intents, but it is not a strong enough signal to classify tickets by itself.

The actual vocabulary and phrases used by customers are more important for distinguishing between intents.

## Q3) can a simple keyword system route tickets?

A simple keyword-based classifier was implemented as a baseline before training the machine-learning model.

The baseline achieved:

- Accuracy: 11.25%
- Coverage: 11.25%

The low coverage means that the predefined rules recognized only a small portion of the test messages.

The baseline worked mainly when customers used explicit phrases that closely matched the manually defined keywords. Messages expressing the same intent using different wording were often missed.

For example, messages such as:

“When will I get my card?” and “How do I track my card?”

both relate to card_arrival, but they may not contain the exact phrases defined in the keyword rules.

- Conclusion:

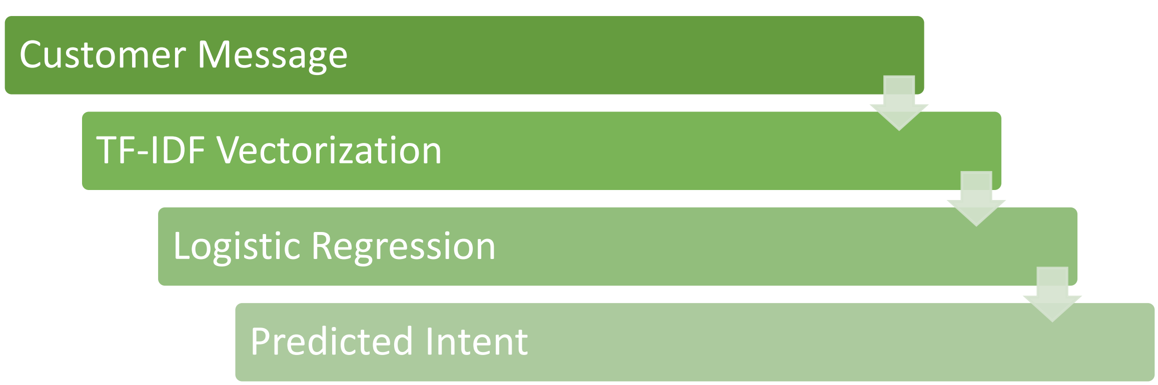
A small manually defined keyword system is not sufficient as the main ticket-routing mechanism for this task.

## Q4) can classical NLP predict customer intent?

Yes. The final system uses TF-IDF combined with Logistic Regression.

TF-IDF converts customer messages into numerical features based on the importance of their terms. Logistic Regression then learns the relationship between these features and the selected customer intents.

The final pipeline is:



The pipeline achieved:

- Accuracy: 97.19%
- Weighted Precision: 97.31%
- Weighted Recall: 97.19%
- Weighted F1-score: 97.18%
- Macro-F1: 97.18%

The results show that classical NLP can perform very well for this constrained eight-intent classification problem.

The use of a scikit-learn Pipeline also ensures that the TF-IDF transformation and classifier are applied consistently during training and prediction.

## Q5) which intents does the model understand well, and which does it confuse?

The model performs strongly across all eight selected intents.

The F1-scores range from 0.938 to 1.000.

The strongest-performing intent is:

- transfer_not_received_by_recipient
  F1-score: 1.000

The lowest-performing intent is:

- card_not_working
  F1-score: 0.938

The model also achieved a Recall of 0.900 for "lost_or_stolen_card", meaning that some actual lost-or-stolen-card cases were classified as other intents.

The model produced 9 incorrect predictions out of 320 test samples.

The confusion matrix shows that several errors occur between semantically related categories. Examples include:

- card_arrival → card_not_working
- lost_or_stolen_card → card_not_working
- lost_or_stolen_card → cash_withdrawal_not_recognised
- cash_withdrawal_charge → cash_withdrawal_not_recognised

These errors suggest that ambiguous or short card-related messages can be more difficult to distinguish, especially when customers mention a card without explicitly describing the underlying problem.

Overall, the model does not appear to rely on only a few easy categories; performance remains consistently high across all eight intents.

## Q6) are all errors equally costly?

No. From a business perspective, different classification errors can have very different consequences.

For this prototype, three intents were considered potentially more business-critical:

- lost_or_stolen_card
- cash_withdrawal_not_recognised
- declined_card_payment

The first two may involve card security or potentially unauthorized activity, while a declined payment can directly prevent a customer from completing a transaction.

For these cases, missing a real ticket may be more problematic than incorrectly routing a less sensitive support request.

Therefore, Recall deserves particular attention for potentially business-critical intents.

A low Recall means that some real cases are missed and assigned to another category. Precision is also important because excessive false positives can increase the workload of support agents.

The classification system should therefore be evaluated using multiple metrics rather than Accuracy alone.

## Q7) how should the system handle a new message?

The trained model is exposed through a reusable prediction function.

When a new customer message arrives, the system:

Receives the message.

Applies the trained TF-IDF transformation.

Predicts the most likely intent using Logistic Regression.

Calculates the model's confidence.

Returns the predicted intent and confidence score.

For example:

Message: “I lost my card”

Predicted intent: “lost_or_stolen_card”

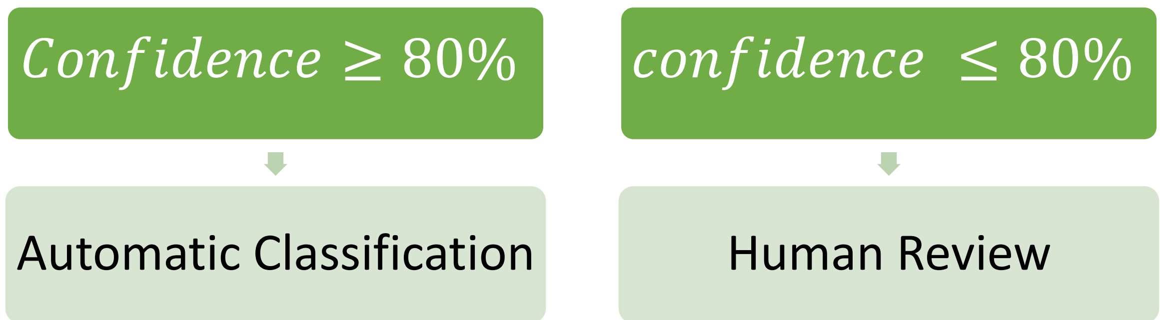
Confidence: “85.01%”

This allows the classifier to be used as an automated routing component for new support tickets rather than only being used for offline evaluation.

## Q8) when should the system defer to a human?

A confidence-based human-review mechanism was introduced.

For this prototype, an initial confidence threshold of 80% is used.




Example predictions:

| Message | Predicted Intent | Confidence | Decision |
|---|---|---|---|
| **I lost my card** | lost_or_stolen_card | 85.01% | Automatic |
| **Why was I charged twice?** | transaction_charged_twice | 90.35% | Automatic |
| **My cash withdrawal is not mine** | cash_withdrawal_not_recognised | 80.16% | Automatic |
| **My card payment was declined** | declined_card_payment | 98.83% | Automatic |

All four example messages exceeded the 80% threshold.

However, the threshold should be considered an initial operational rule, not a guarantee of correctness. In particular, the “cash_withdrawal_not_recognised” example received only 80.16%, which is very close to the threshold.

For a production deployment, the threshold should be calibrated using validation data and the business cost of different types of errors. Higher-risk intents could also be assigned stricter review requirements.

## Overall Recommendation

The results indicate that the TF-IDF + Logistic Regression system is a strong candidate for automated routing of routine customer-support messages within the selected eight intents.

The model achieved approximately 97.2% accuracy and 97.2% Macro-F1, while producing only 9 errors out of 320 test samples.

However, the system should not be treated as a fully autonomous production solution. A confidence-based human-review mechanism is recommended, particularly for uncertain predictions and potentially high-risk cases such as lost or stolen cards and unrecognized cash withdrawals.

The current system is therefore best viewed as a high-performing prototype for automated ticket classification with human oversight.